DEPARTMENT OF CSE

CS T61 ENTERPRISE SOLUTIONS

Unit-I – ERP AND ITS RELATED TECHNOLOGIES

Introduction : ERP - Definition – Concept – Fundamentals – Need for ERP - Advantages of ERP – Implementation of ERP – Key issues and Characteristics of ERP Typical architecture components of ERP – ERP system Architecture. **ERP and related technologies**: Business Process RE-engineering – Management Information System– Decision Support System - Executive Support System – On-Line Analytical Processing, Supply Chain Management, Customer Relationship Management.

PART A

1. **Define ERP.**
   - Enterprise Resource Planning (ERP) is business management software.
   - It typically a suite of integrated applications that a company can use to collect, store, manage and interpret data from many business activities, including product planning, cost , Manufacturing or service delivery, Marketing and sales, Inventory.
2. **What is enterprise solutions?**
   - Enterprise Solutions provide for a scalable, easy to manage programming solution to providing business management and information accessibility for internal and external clients.
   - Enterprise solutions deal with the problem of providing information to clients both externally and internally. It deals with programming and databases.
   - The main problem being how to most efficiently get our data available to those we want to access it. The solution have the following characteristics and more: Security, Scalability, Cost, and Management Portable.
3. **Describe about Business Intelligence.**
   - Business Intelligence (BI) has become a standard component of most ERP packages.
   - In general, BI tools allow user to share and analyze the data collected the enterprise and centralized in the ERP database.
   - BI can come in the form of dashboards, automated reporting and analysisis tools used to monitor the organization as business performance. BI supports informed decision making by everyone, from executives to line managers and accountants.

4. **List out the Characteristics of ERP.**

The characteristics of ERP are

- Adaptability
- Openness
- Integration
- Completeness
- Homogenization
- Transversality
- Best practices
- Real time
- Simulation

5. **What is the need of ERP?**

- Separate systems were being maintained during 1960/70 for traditional business functions like Sales & Marketing, Finance, Human Resources, Manufacturing, and Supply Chain Management.
- These systems were often incongruent, hosted in different databases and required batch updates. It was difficult to manage business processes across business functions e.g. procurement to pay and sales to cash functions.
- ERP system grew to replace the islands of information by integrating these traditional business functions.

6. **List the advantages of ERP.**

- Complete **visibility** into all the important processes, across various departments of an organization (especially for senior management personnel).
- Automatic and coherent **workflow** from one department/function to another, to ensure a smooth transition and quicker completion of processes. This also ensures that all the inter-departmental activities are properly tracked and none of them is 'missed out'.
- A unified and single **reporting** system to analyze the statistics/status etc. in real-time, across all functions/departments.

7. **What are the fundamental components of ERP?**

The following are the fundamental components of the ERP.

- Financial Management
- Business Intelligence
- Supply Chain Management

- Human Resource Management
- Manufacturing Operations
- Integration

8. List out the elements of ERP.

The following are the elements of ERP.

- Production planning
- Integrated logistics
- Human resources
- Accounting and financials
- Sales, distribution (order entry)

9. Why companies are forced to have ERP suite?

The forces driving ERP are;

- The need to create a framework that will, improve customer order processing.
- The need to consolidate and unify business functions such as manufacturing finance distributions/logistics, & human resources.
- The need to integrate a broad range of desperate technologies, along with the processes they support.
- The need to create a new foundations and which next-generation applications can be developed.

10. In what way ERP is used in real world?

- Microsoft in the software industry
- Owens-Corning in the building supplies industry.
- Colgate-Palmolive in the consumer products industry

11. List the implementation methodology phases of accelerated ERP approach.

The Project Preparation Phase

- The Blueprint Phase
- The Pilot Phase
- The Assessment Phase
- The Final Phase

12. What are the capabilities of effective co-ordination management?

The capabilities of effective co-ordination managements are,

- Strategic Thinking
- Process Reengineering
- Managing Implementation Complexity
- Transition Management

13. Mention the elements required in ERP to achieve flexibility.

Four crucial elements are required to achieve flexibility:

1. Components, not modules.
2. Incremental migration, rather than massive reengineering
3. Dynamic, rather than static, configuration of ERP systems.
4. Management of multiple strategic sourcing and partnership relationships.

14. What is Two Tier architecture?

- In typical two-tier architecture, the server handles both application and database duties.
- The clients are responsible for presenting the data and passing user input back to the server.
- While there may be multiple servers and the clients may be distributed across several types of local and wide area links, this distribution of processing responsibilities remains the same.

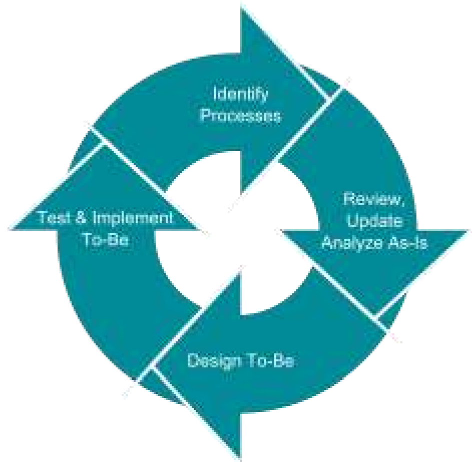
15. What is Business Process Reengineering (BPR)?

- Business process re-engineering: Business process re-engineering (BPR) is the analysis and redesign of workflows within and between enterprises in order to optimize end-to-end processes and automate non-value-added tasks.
- Business Process Reengineering (BPR) is the practice of rethinking and redesigning the way work is done to better support an organization's mission and reduce costs.
- Reengineering starts with a high-level assessment of the organization's mission, strategic goals, and customer needs.

16. Define Management Information System.

A management information system (MIS) is a computerized database of financial information organized and programmed in such a way that it produces regular reports on operations for every level of management in a company.

17. Depict the neat diagram of Business Process Reengineering cycle.



Business Process Reengineering Cycle

18. List out the advantages of BPR.

Advantages of BPR are

- Satisfactions
- Growth of knowledge
- Solidarity of the company
- Demanding jobs
- Authority

19. Write a note on Decision Support System (DSS).

- Decision support system (DSS) is a computer-based information system that supports business or organizational decision-making activities.
- DSSs serve the management, operations, and planning levels of an organization (usually mid and higher management) and help people make decisions about problems that may be rapidly changing and not easily specified in advance.
- Unstructured and Semi-Structured decision problems.
- Decision support systems can be either fully computerized, human-powered or a combination of both.

20. What are the various phases are available in DSS.

Decision Support System consists of four phases:

- **Intelligence**: Searching for conditions that call for decision.
- **Design**: Developing and analyzing possible alternative actions of solution.
- **Choice**: Selecting a course of action among those.
- **Implementation**: Adopting the selected course of action in decision situation.

21. Define Executive Support System.

- An executive information system (EIS), also known as an executive support system (ESS), is a type of management information system that facilitates and supports senior executive information and decision-making needs.
- It provides easy access to internal and external information relevant to organizational goals.
- It is commonly considered a specialized form of decision support system (DSS).
- EIS emphasizes graphical displays and easy-to-use user interfaces.

22. What is On-Line Analytical Processing (OLAP)?

- OLAP (Online Analytical Processing) is the technology behind many Business Intelligence (BI) applications.
- OLAP is a powerful technology for data discovery, including capabilities for limitless report viewing, complex analytical calculations, and predictive "what if" scenario (budget, forecast) planning.

23. Define Multidimensional OLAP.

- MOLAP (multi-dimensional online analytical processing) is the classic form of OLAP and is sometimes referred to as just OLAP.
- MOLAP stores this data in an optimized multi-dimensional array storage, rather than in a relational database.

24. Mention the advantages of Multidimensional OLAP.

- Fast query performance due to optimized storage, multidimensional indexing and caching.
- Smaller on-disk size of data compared to data stored in relational database due to compression techniques.
- Automated computation of higher level aggregates of the data.
- It is very compact for low dimension data sets.
- Array models provide natural indexing.
- Effective data extraction achieved through the pre-structuring of aggregated data.

25. Draft a note on Supply Chain Management (SCM).

- Supply chain management (SCM) is the management of the flow of goods and services.
- It includes the movement and storage of raw materials, work-in-process inventory, and finished goods from point of origin to point of consumption.

26. Define Customer Relationship Management (CRM).

Customer relationship management (CRM) is an approach to managing a company's interaction with current and future customers. It often involves using technology to organize, automate, and synchronize sales, marketing, customer service, and technical support.

27. What are the characteristic of CRM?

- Sales force automation, which implements sales promotion analysis, automates the tracking of a client's account history for repeated sales or future sales, and coordinates sales, marketing, call centers, and retail outlets.
- Data warehouse technology, used to aggregate transaction information, to merge the information with CRM products, and to provide key performance indicators.
- Opportunity management which helps the company to manage unpredictable growth and demand, and implement a good forecasting model to integrate sales history with sales.

28. Differentiate B2C and B2B.

- B2B companies have smaller contact databases than B2C.
- The volume of sales in B2B is relatively small.
- In B2B there are less figure propositions, but in some cases they cost a lot more than B2C items.
- Relationships in B2B environment are built over a longer period of time.

29. Explain ROLAP.

- ROLAP – Relational OLAP works directly with relational databases.
- The base data and the dimension tables are stored as relational tables and new tables are created to hold the aggregated information.
- It depends on a specialized schema design.
- This methodology relies on manipulating the data stored in the relational database to give the appearance of traditional OLAP's slicing and dicing functionality.

30. Name some of the applications of EIS.

Applications of EIS are,

- Manufacturing
- Marketing
- Financial

PART – B (11 MARKS)

1. Discuss about ERP and its Functional Components.

DEFINITION OF ERP:

- Enterprise Resource Planning (ERP) is business management software.
- It typically a suite of integrated applications that a company can use to collect, store, manage and interpret data from many business activities, including product planning, cost , Manufacturing or service delivery, Marketing and sales, Inventory .
- An important goal of ERP is to facilitate the flow of information so business decisions can be data-driven.
- ERP software suites are built to collect and organize data from various levels of an organization to provide management with insight into key performance indicators (KPIs) in real time.
- ERP is the backbone of e-business.
- ERP integrates application suites.
- ERP is not a single system, but a framework that includes administrative apps (finance, accounting), human resource apps (payroll, benefits), and manufacturing resource planning (MRP) apps (procurement, production planning). ERP unites major business processes – order processing, general ledger, payroll, and production within a single family of software modules.

Fundamental Components of ERP:

The following are the fundamental components of the ERP.

1. Financial Management
2. Business Intelligence
3. Supply Chain Management
4. Human Resource Management
5. Manufacturing Operations
6. Integration

Financial Management:

- At the core of ERP are the financial modules, including general ledger, accounts receivable, accounts payable, billing and fixed asset management.
- If organization is considering the move to an ERP system to support expansion into global markets, make sure that multiple currencies and languages are supported, as well as regulatory compliance in the U.S. and in foreign countries.

- Other functionality in the financial management modules will include budgets, cash-flow, expense and tax reporting.
- The evaluation team should focus on areas that are most important to support the strategic plans for your organization.

Business Intelligence:

- Business Intelligence (BI) has become a standard component of most ERP packages.
- In general, BI tools allow users to share and analyze the data collected across the enterprise and centralized in the ERP database.
- BI can come in the form of dashboards, automated reporting and analysis tools used to monitor the organization business performance.
- BI supports informed decision making by everyone, from executives to line managers and accountants.

Supply Chain Management:

- Supply Chain Management (SCM), sometimes referred to as logistics, improves the flow of materials through an organization by managing planning, scheduling, procurement, and fulfillment, to maximize customer satisfaction and profitability.
- Sub modules in SCM often include production scheduling, demand management, distribution management, inventory management, warehouse management, and procurement and order management.
- Any company dealing with products, from manufacturers to distributors, needs to clearly define their SCM requirements to properly evaluate an ERP solution.
- It easy for a vendor to focus on their applications strengths and not address the full needs of the company.

Human Resource Management:

- Human resource management ERP modules should enhance the employee experience – from initial recruitment to time tracking.
- A Sub module can include payroll, performance management, time tracking, benefits, compensation and workforce planning.
- Self-service tools that allow managers and employees to enter time and attendance, choose benefits and manage PTO are available in many ERP solutions.

Manufacturing Operations:

- Manufacturing modules make manufacturing operations more efficient through product configuration, job costing and bill of materials management.
- ERP manufacturing modules often include Capacity Requirements Planning, Materials Requirements Planning, forecasting, Master Production Scheduling, work-order management and shop-floor control.

Integration:

- Key to the value of an ERP package is the integration between modules, so that all of the core business functions are connected.
- Information should flow across the organization so that BI reports on organization-wide results.
- ERP can be easier than you imagine – Microsoft Dynamics ERP is cost effective and familiar to your users. If you are thinking about upgrading your systems to a fully integrated ERP system, give us a call.

NEED FOR ERP:

1. Real-time information for decisions
2. Best practice procedures
3. Improved visibility
4. Faster month-end close
5. Increased customer satisfaction
6. Managed and controlled costs
7. Better operational efficiency
8. Accurate records
9. Balance of supply and demand
10. Reduced lead times and increased throughput.

Real-time information for decisions:

- Without an ERP system, team is flying blind.
- They make decisions based on guesswork and rules of thumb because they don't have the data they need.
- Sometimes they are the right decisions, but more often, they are sub-optimum decisions that can cost you money and customer goodwill.

Best practice procedures:

- Software companies often design their ERP systems to support specific industries or verticals.
- As they add customers, they learn industry best practices and incorporate them into the software.
- By implementing an ERP system designed for your industry, you automatically make your business processes more efficient.

Improved visibility:

- If customers want to know when their order will ship or if you need to know whether you have enough of a critical component to accept a rush order, an ERP system gives you instant visibility into your operations and your supply chain.

Faster month-end close:

- ERP systems automatically process transactions and generate audit trails and financial reports that can simplify period-end closings.
- They flag anomalies so you can investigate quickly, and they simplify repetitive journal entries and other activities that make closing so complex and time consuming. Faster closes mean you know the health of your business sooner.

Increased customer satisfaction:

- Customers like accurate delivery dates, and ERP can help you provide them with improved inventory and shop floor visibility.
- In addition, the increased visibility and accuracy will help you improve your DIFOT rate (delivery in full on time), which helps keep customers happy.

Managed and controlled costs:

- ERP systems calculate and collect costs so you always have an accurate picture of your product cost and margins.

Better operational efficiency:

- By helping you to plan production more effectively, your operational efficiency will improve as you reduce setups and teardowns or unnecessary downtime.

Accurate records:

- The uniformity of record data that an ERP system instills will help ensure that your records are more accurate, which will increase process accuracy across the board.

Balance of supply and demand:

- MRP, a component of ERP systems, will help you balance supply and demand so you can reduce inventory while keeping customers happy.

Reduced lead times and increased throughput:

- Better scheduling and accurate records ensure that your schedules focus on priorities, leading to shorter lead times. Since you won't have as many orders waiting for tooling or parts, your throughput will increase.

2. **Explain in detail about Implementation of ERP, key issues and its characteristics.**

Implementation of ERP

- Implementation of ERP System, is a complex exercise, involving many process alterations and several legacy issues.
- Organizations need an implementation strategy encompassing both pre implementation and implementation stages. The fallout of a poor strategy is unpreparedness of employees, implementation not in conformity with wider business strategy, poor business process redesign and time and cost overrun.
- Following issues must be carefully thought out and formulated, as a part of implementation strategy, before embarking on actual implementation:

Business Process:

- Hypothetically, company insiders should know best about the processes of their organization. But employees often constrained to work in departmental silos and overlook wood for the tree.
- Under most circumstances, prevailing business practices are not properly defined and no "as is" flow charts, documenting existing processes, are available.
- An ERP implementation could be a great occasion to assess and optimize existing business processes, control points, breaking points between departments, and interfaces with trading partners.
- But, often, due to resistance to changes and departmental clouts, ERP implementation is comprehended as an exercise to automate legacy processes.
- This may lead to little improvement in underlying business processes, resulting no appreciable return on investment.
- Automating existing manual processes peculiar to a company necessitates, significant source code customization, as even a best fit ERP product match to a maximum of 85% to 90% of legacy processes.

- Source code customization will not only require changing of software objects but also need changing data models. The efforts needed to make such changes are significant in terms of development, testing and documentation.
- The future cost of maintenance and upgrades will be substantial, affecting entire life cycle of the system.
- Unless a considered view favoring process changes is taken as a part of implementation strategy, pressure will mount subsequently for more and more customization, when the exercise of Business Process Mapping and Gap Analysis is taken up during implementation.
- ERP systems are highly configurable and contain series of design trade off to meet various nuances of the same business cycles / processes.
- This should, normally, be sufficed to cover needed processes, probably with a little bit of swapping whenever needed.
- At occasions, it may be imperative to change source code to account for some unique core processes of the organizations.
- Procedure for authorization of such changes, normally requiring attention from sponsor, should also form part of the strategy document.

Implementation Methodology:

- Selection of implementation methodology constitutes an important component of implementation strategy. Most popular implementation methodology is "big bang" approach where on a scheduled cut-off date; entire system is installed throughout the organization.
- All users move to the new system and manual / legacy systems are discontinued. The implementation is swift and price tag is lesser than a phased implementation.
- On the flip side, risk element is much higher and resources for training, testing and hand holding are needed at a much higher level, albeit for a shorter period of time
- Another major implementation strategy is "phased implementation", where roll out is done over a period. This method is less focused, prolonged and necessitates maintenance of legacy System over a period of time.
- But, phased implementation is less risky, provides time for user's acquaintance and fall back scenarios are less complicated.

  There are various choice of phasing such as

  - Phased roll out by locations for a multi-location company
  - Phased roll out by business unit e.g. human resources

- Phased roll out by module e.g. general ledger.

Implementation methodology is depends on several factors

- Size
- Industry
- Sales volume

The Common basic factors of ERP is

- Physical Scope, BPR, Resource Allocation

The 3 Broad implementation strategies are:

1. Big Bang
2. Middle-road
3. Vanilla

1. BIG BANG APPROACH

- What is the “Big Bang” approach?

  A straightforward ERP Implementation approach. It means all ERP modules, such as financials, manufacturing, and human resources, etc., are implemented in all business units’ at all geographic locations at the same time.
- It will push the entire organization to use the new system at the same time.
- The old system will be entirely shut down

IMPLEMENTATION PROCESS

- Converting the system.

  Planning, convert data from old system, load data in new system, test data in new system, execute off-line trials, and check to verify validity.
- Releasing parts of the system.

  Release converted database, release produced application, release infrastructure.
- Training the future users.

  Create main buffer of experienced staff, training all users.
- Releasing the whole system.
- Turn down the old system and load the new system.

Middle Road Approach

- Not all ERP modules are implemented
- Industry specific modules and sub modules can be chosen
- Some legacy systems to remain functional.

Vanilla Approach

- Only essential modules are chosen
- Industry specific modules are discouraged
- Minimized risk

Challenges in implementing ERP solutions are quite normal. Though it is not completely a technical job, a lot of planning and proper communication is very much essential to implement ERP across the organization.

Below are the 7 common challenges we have noticed companies experience, when ERP is implemented.

- It is very important, that **implementation is done in stages**. Trying to implement everything at once will lead to a lot of confusion and chaos.
- **Appropriate training is very essential** during and after the implementation. The staff should be comfortable in using the application or else, it will backfire, with redundant work and functional inefficiencies.
- **Lack of proper analysis of requirements** will lead to non-availability of certain essential functionalities. This might affect the operations in the long run and reduce the productivity and profitability.
- **Lack of Support from Senior Management** will lead to unnecessary frustrations in work place. Also, it will cause delay in operations and ineffective decisions. So, it is essential to ensure that the Senior Management supports the transformation.
- **Compatibility Issues with ERP Modules** lead to issues in integration of modules. Companies associate different vendors to implement different ERP modules, based on their competency. It is very essential that there is a way to handle compatibility issues.
- **Cost Overheads** will result, if requirements are not properly discussed and decided during the planning phase. So, before execution, a detailed plan with a complete breakdown of requirements should be worked out.

- **Investment in Infrastructure** is very essential. ERP applications modules will require good processing speed and adequate storage. Not allocating suitable budget for infrastructure will result in reduced application speed and other software issues. Hardware and Software Security is also equally important.

Characteristics of ERP

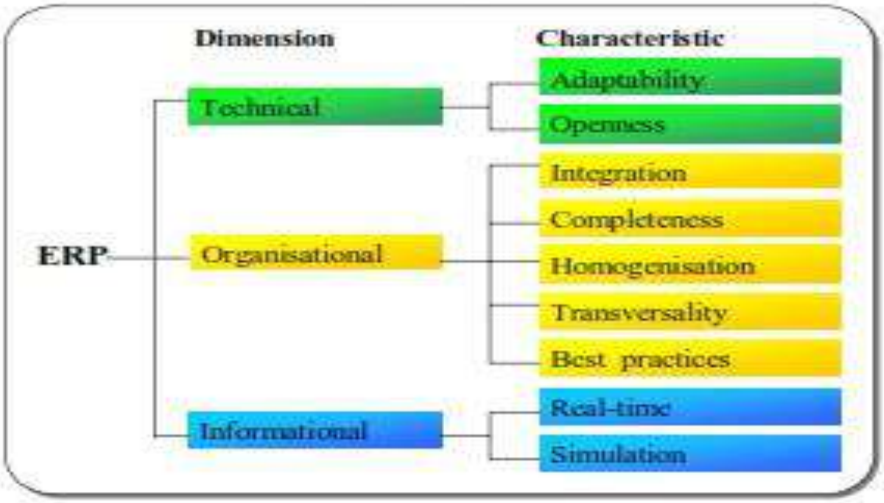
- ERP (Enterprise Resource Planning) systems typically include the following characteristics:
- An integrated system that operates in (or near) real time without relying on periodic updates
- A common database that supports all applications
- A consistent look and feel across modules
- Installation of the system with elaborate application/data integration by the Information Technology department provided the implementation is not done in small steps.
- Flexible
- Modular and Open
- Comprehensive
- Online-Connection with Other ERP System
- Best Business Practices
- Multi-Facilities
- Strategic planning
- Optimize the data
- Project Management
- Automatic Functions

Key issues

Legacy data: Gathering, cleaning and removing of duplicate data.

Hardware and software: Addition and updating of existing resources. Compatibility with existing Operating system and Database

Project structure: Project champions and competency center



Adaptability

- ERP system was not adaptable, it would limit the organization's development potential in a changing environment.
- Also, given that the ERP system integrates various functions of the organization (Production, finance, HRM, etc.), there seems to be both an opposing and a complementary.

Openness

- Openness is a characteristic that appears to be redundant in regard to flexibility.
- The modularity and portability (openness) that allow an ERP system to evolve with the organization can be considered as factors of flexibility.

Integration

- Integration is without a doubt the most important ERP characteristics.
- It distinguishes ERP systems from traditional IS whose informational fragmentation has been criticized in that it reflects a vision of the organization as a set of islands or functional silos that cannot communicate with each other, or that communicate little or with great difficulty.

Completeness

- Completeness (the generic functionality) is a characteristic that, pushed to the extreme, becomes unrealistic.
- A generic system that would work for all types of firms and industries is in fact very difficult if not impossible to design.

Homogenization

- Homogenization refers to the existence of a unique data referential, the uniformity of human-machine interfaces, and to the unicity of the application system's administration.

Transversality

- Transversality refers to the process-oriented view of an ERP system.
- Its importance comes from the fact that ERP systems are composed of functional modules, and are generally implemented module by module, that is, in a vertical manner.

Best Practices

- Imbedding best practices would not be considered as an essential characteristic of ERP systems.
- The reason is that this notion is based on adopting generic processes that, despite being exemplary, offer few possibilities of gaining a competitive advantage.

Real Time

- The capacity for real-time operation and simulation in an ERP system are consequences of its successful integration.
- It is this integration that allows the same information to be communicated in real time to all parts of the organization, and allows simulating the effect of an input on all activities of the organization

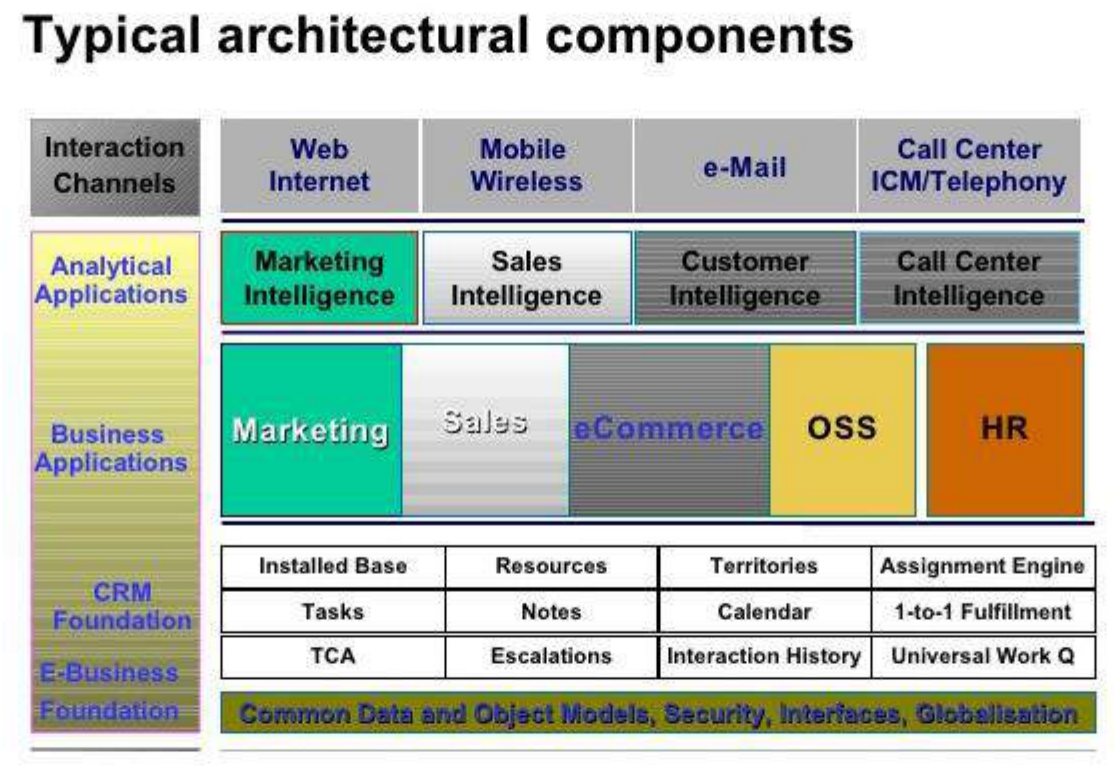
CHARACTERISTICS OF TYPICAL ARCHITECTURE COMPONENTS OF ERP

Integration

- Seamless integration of all the information flowing through a company – financial and accounting, human resource information, supply chain information, and customer information.

Packages

- Enterprise systems are not developed in-house
- IS life cycle is different
    1. Mapping organizational requirements to the processes and terminology employed by the vendor and
    2. Making informed choices about the parameter setting.
- Organizations that purchase enterprise systems enter into long-term relationships with vendors. Organizations no longer control their own destiny.



Best Practices

- ERP vendors talk to many different businesses within a given industry as well as academics to determine the best and most efficient way of accounting for various transactions and managing different processes. The result is claimed to be "industry best practices".
- The general consensus is that business process change adds considerably to the expense and risk of an enterprise systems implementation. Some organizations rebel against the inflexibility of these imposed business practices.

Evolving

- Enterprise Systems are changing rapidly
- Architecturally: Mainframe, Client/Server, Web-enabled, Object-oriented, Componentization
- Functionally: front-office (i.e. sales management), supply chain (advanced planning and scheduling), data warehousing, specialized vertical industry solutions, etc.

3. Sketch and explain about ERP System Architecture.

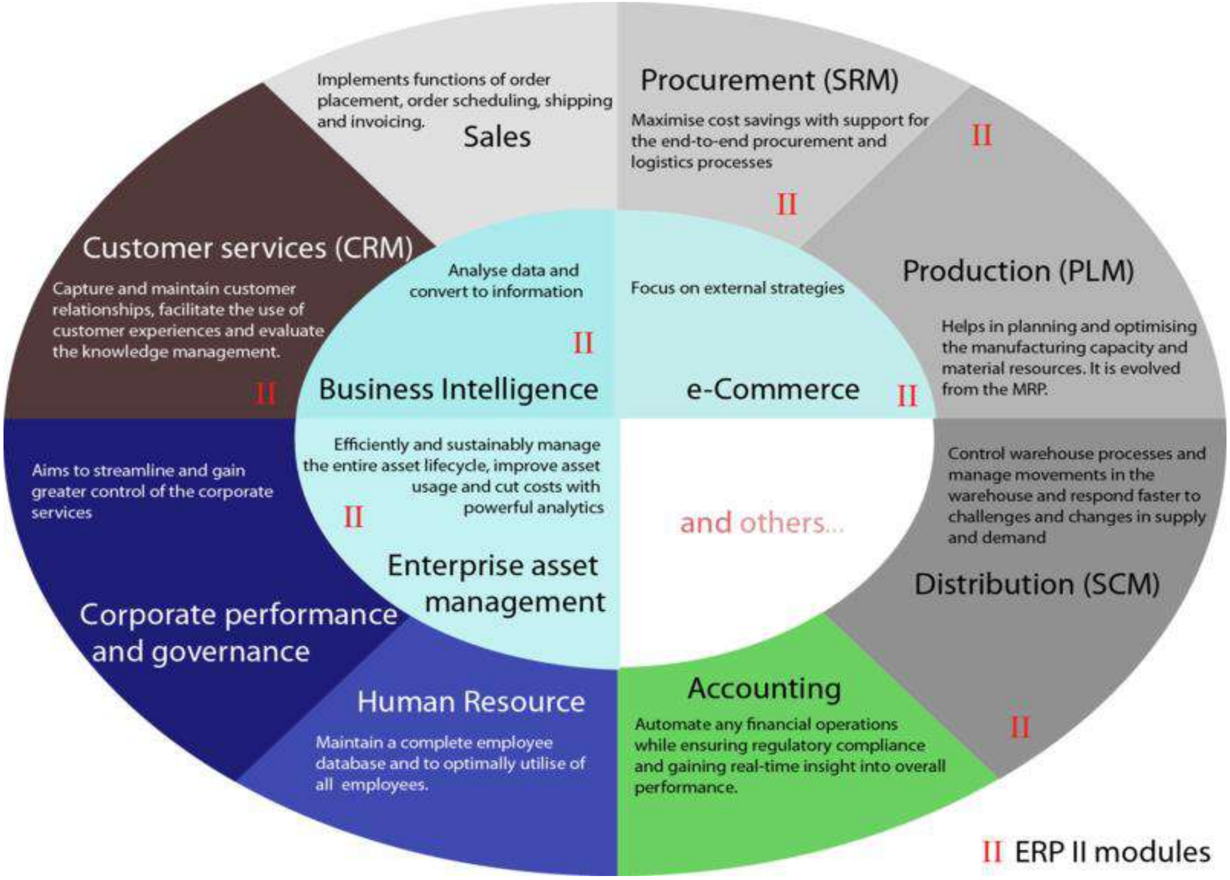
ERP SYSTEM ARCHITECTURE:

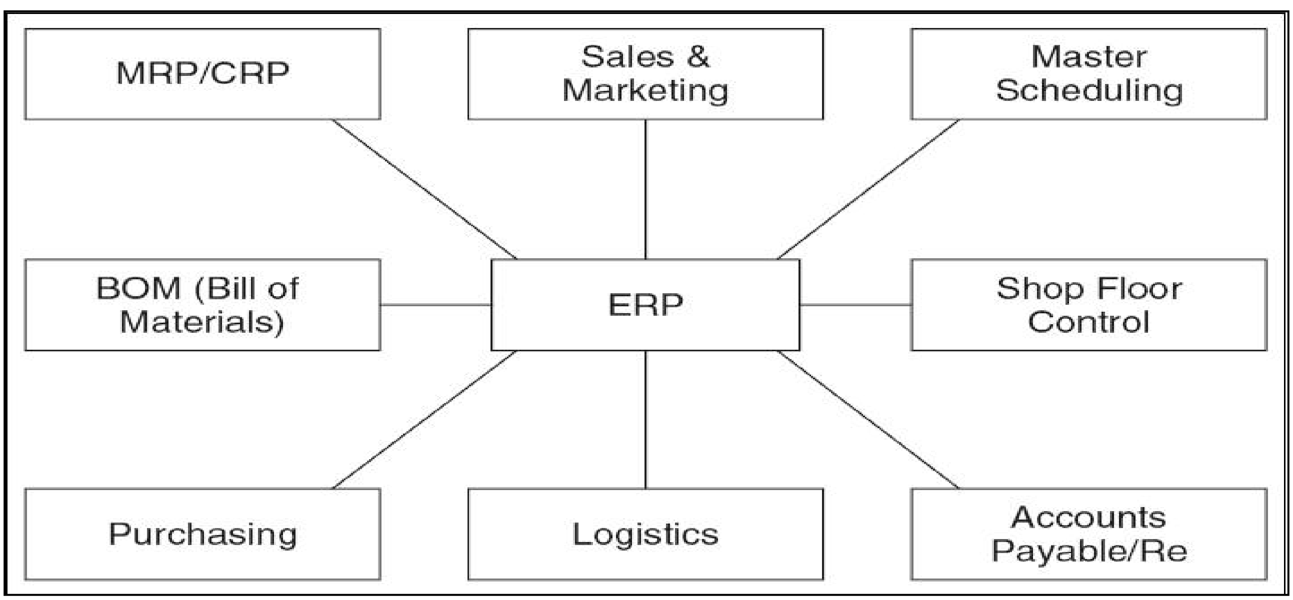
- Enterprise resource planning (ERP) is business-management software typically a suite of integrated applications that an organization can use to collect, store, manage and interpret data from many business activities, including:
    - Product planning, cost
    - Manufacturing or service delivery
    - Marketing and sales
    - Inventory management
    - Shipping and payment
- ERP provides an integrated view of core business processes, often in real-time, using common databases maintained by a database management system.
- ERP systems track business resources such as cash, raw materials, production capacity and the status of business commitments: orders, purchase orders, and payroll.
- The applications that make up the system share data across various departments (manufacturing, purchasing, sales, accounting, etc.) that provide the data. ERP facilitates information flow between all business functions, and manages connections to outside stakeholders.
- The ERP system is considered a vital organizational tool because it integrates varied organizational systems and facilitates error-free transactions and production. However, developing an ERP system differs from traditional system development.
- ERP systems run on a variety of computer hardware and network configurations, typically using a database as an information repository

ERP modules:

- The key role of an ERP system is to provide support for such business functions as accounting, sales, inventory control, and production.
- ERP vendors, including SAP, Oracle, and Microsoft, etc. provide modules that support the major functional areas of a business.
- The ERP software embeds best business practices that implement the organization's policy and procedure via business rules

Typical ERP Modules:





- ERP applications are most commonly deployed in a distributed and often widely dispersed manner. While the servers may be centralized, the clients are usually spread to multiple locations throughout the enterprise.

- Generally there are three functional areas of responsibility that is distributed among the servers and the clients.
- First, there is the database component - the central repository for all of the data that is transferred to and from the clients.
- Then, of course, the clients - here raw data gets inputted, requests for information are submitted, and the data satisfying these requests is presented.
- Lastly, we have the application component that acts as the intermediary between the client and the database. Where these components physically reside and how the processes get distributed will vary somewhat from one implementation to the next.

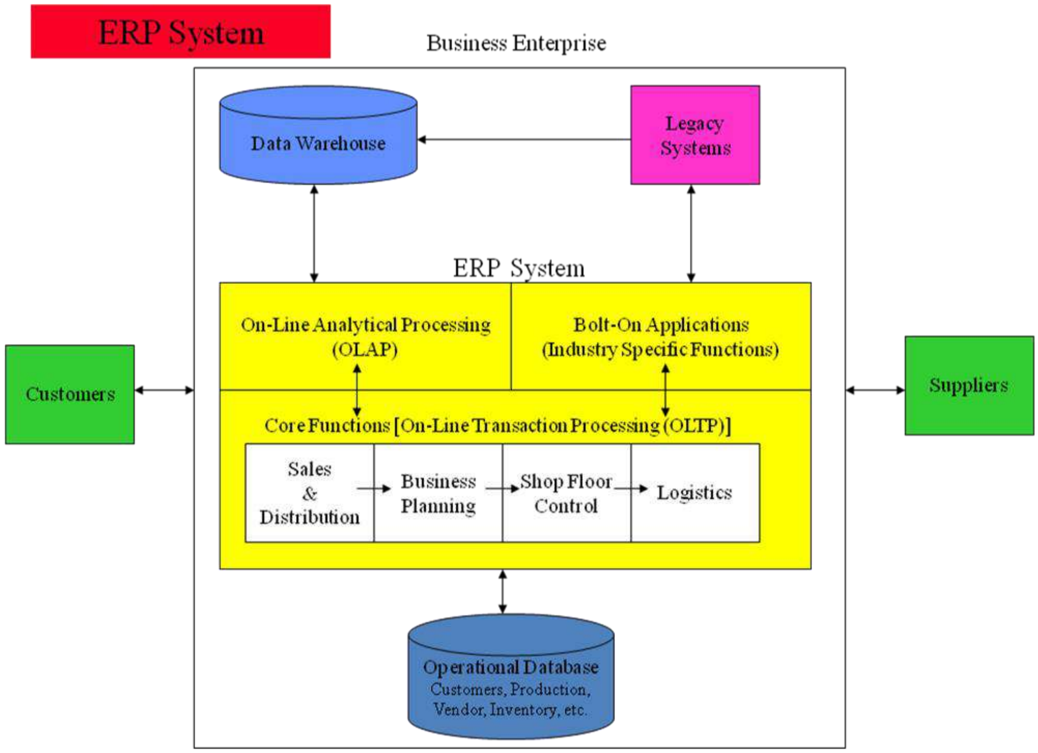
The two most commonly implemented architectures are outlined below.

Two-tier Implementations

- In typical two-tier architecture, the server handles both application and database duties. The clients are responsible for presenting the data and passing user input back to the server.
- While there may be multiple servers and the clients may be distributed across several types of local and wide area links, this distribution of processing responsibilities remains the same.

Three-tier Client/Server Implementations

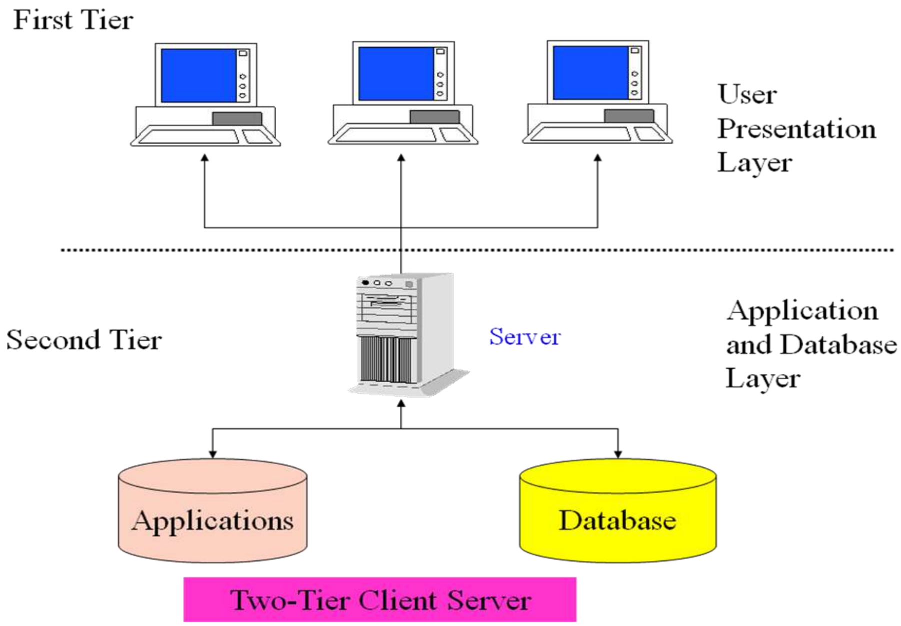
- In three-tier architectures, the database and application functions are separated. This is very typical of large production ERP deployments.
- In this scenario, satisfying client requests requires two or more network connections. Initially, the client establishes communications with the application server. The application server then creates a second connection to the database server.



ERP System Configurations: Client-Server Network Topology

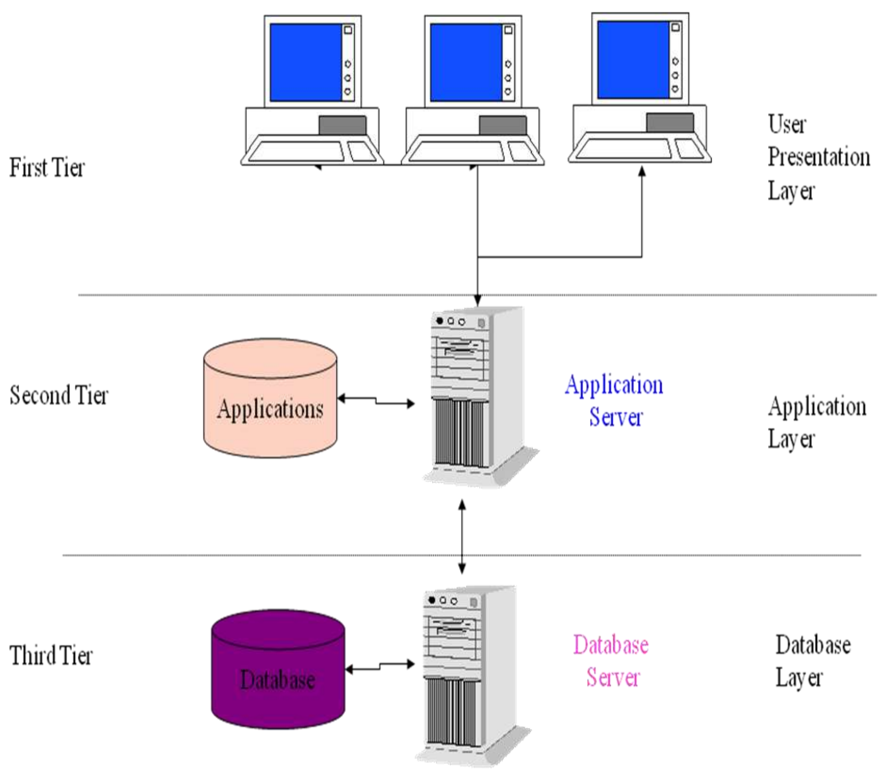
Two-tier:

- It common server handles both application and database duties.
- It is used especially in LANs.



Three-tier:

- The client links to the application server which then initiates a second connection to the database server.
- It is used especially in WANs



Three-Tier Client Server

4. Discuss the advantages and disadvantages of ERP.

<u>ADVANTAGES</u>

- Complete **visibility** into all the important processes, across various departments of an organization (especially for senior management personnel).
- Automatic and coherent **workflow** from one department/function to another, to ensure a smooth transition and quicker completion of processes. This also ensures that all the inter-departmental activities are properly tracked and none of them is 'missed out'.
- A unified and single **reporting** system to analyze the statistics/status etc. in real-time, across all functions/departments.
- Since **same (ERP) software** is now used across all departments, individual departments having to buy and maintain their own software systems is no longer necessary.

- Certain ERP vendors can extend their ERP systems to provide Business **Intelligence** functionalities that can give overall insights on business processes and identify potential areas of problems/improvements.
- Advanced **e-commerce integration** is possible with ERP systems – most of them can handle web-based order tracking/ processing.
- There are **various modules** in an ERP system like Finance/Accounts, Human Resource Management, Manufacturing, Marketing/Sales, Supply Chain/Warehouse Management, CRM, Project Management, etc.
- Since ERP is a **modular software** system, it's possible to implement either a few modules (or) many modules based on the requirements of an organization. If more modules implemented, the integration between various departments may be better.
- Since a Database system is implemented on the backend to store all the information required by the ERP system, it enables **centralized storage/back-up** of all enterprise data.
- ERP systems are **more secure** as centralized security policies can be applied to them. All the transactions happening via the ERP systems can be tracked.
- ERP systems provide better company-wide visibility and hence enable **better/faster collaboration** across all the departments. 12. It is possible to integrate other systems (like bar-code reader, for example) to the ERP system through an API (Application Programing Interface). 13. ERP systems make it easier for order tracking, inventory tracking, revenue tracking, sales forecasting and related activities.
- ERP systems are especially helpful for managing globally dispersed enterprise companies, better.

DISADVANTAGES

- The cost of ERP Software, planning, customization, configuration, testing, implementation, etc. is too high.
- ERP deployments are highly time-consuming – projects may take 1-3 years (or more) to get completed and fully functional.
- Too little **customization** may not integrate the ERP system with the business process & too much customization may slow down the project and make it difficult to upgrade.
- The cost savings/payback may not be realized immediately after the ERP implementation & it is quite difficult to measure the same.

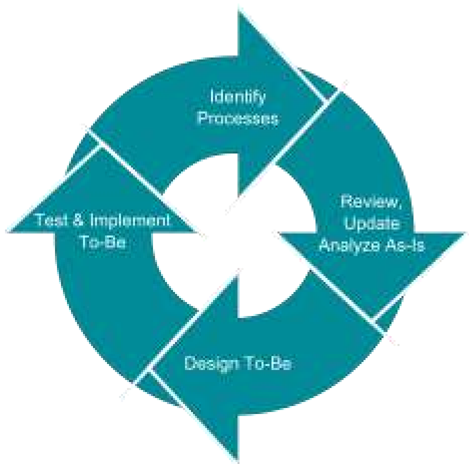
- The **participation** of users is very important for successful implementation of ERP projects – hence, exhaustive user training and simple user interface might be critical. But ERP systems are generally difficult to learn (and use).
- There may be **additional indirect** costs due to ERP implementation – like new IT infrastructure, upgrading the WAN links, etc.
- **Migration** of existing data to the new ERP systems is difficult (or impossible) to achieve. Integrating ERP systems with other standalone software systems is equally difficult (if possible). These activities may consume a lot of time, money & resources, if attempted.
- ERP implementations are difficult to achieve in **decentralized organizations** with disparate business processes and systems.
- Once an ERP systems is implemented it becomes a **single vendor lock-in** for further upgrades, customizations etc. Companies are at the discretion of a single vendor and may not be able to negotiate effectively for their services.
- **Evaluation** prior to implementation of ERP system is critical. If this step is not done properly and experienced technical/business resources are not available while evaluating, ERP implementations can (and have) become a failure.

5. Explain in detail about Business Process Re-engineering (BPR).

Business Process Re-engineering (BPR).

- Business process re-engineering (BPR) is the analysis and redesign of workflows within and between enterprises in order to optimize end-to-end processes and automate non-value-added tasks.
- BPR is the practice of rethinking and redesigning the way work is done to better support an organization's mission and reduce costs.
- Reengineering starts with a high-level assessment of the organization's mission, strategic goals, and customer needs.
- Within the framework of this basic assessment of mission and goals, re-engineering focuses on the organization's business processes—the steps and procedures that govern how resources are used to create products and services that meet the needs of particular customers or markets.
- As a structured ordering of work steps across time and place, a business process can be decomposed into specific activities, measured, modeled, and improved.
- It can also be completely redesigned or eliminated altogether. Re-engineering identifies, analyzes, and re-designs an organization's core business processes with the aim of achieving dramatic improvements in critical performance measures, such as cost, quality, service, and speed.

- Re-engineering recognizes that an organization's business processes are usually fragmented into sub processes and tasks that are carried out by several specialized functional areas within the organization.
- Re-engineering maintains that optimizing the performance of sub processes can result in some benefits, but cannot yield dramatic improvements if the process itself is fundamentally inefficient and outmoded.
- For that reason, re-engineering focuses on re-designing the process as a whole in order to achieve the greatest possible benefits to the organization and their customers.



Business Process Reengineering Cycle

The most notable definitions of reengineering are:

- The fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical contemporary modern measures of performance, such as cost, quality, service, and speed.
- Encompasses the envisioning of new work strategies, the actual process design activity, and the implementation of the change in all its complex technological, human, and organizational dimensions."

Model based on PRLC approach:

- Although the labels and steps differ slightly, the early methodologies that were rooted in IT-centric BPR solutions share many of the same basic principles and elements.
- The following outline is one such model, based on the PRLC (Process Reengineering Life Cycle). Simplified schematic outline of using a business process approach, exemplified for pharmaceutical R&D.

- Structural organization with functional units
- Introduction of New Product Development as cross-functional process
- Re-structuring and streamlining activities, removal of non-value adding tasks

BPR Success & Failure Factors:

- Factors that are important to BPR success include:
- BPR team composition.
- Business needs analysis.
- Adequate IT infrastructure.
- Effective change management.
- Ongoing continuous improvement

There are many reasons for sub-optimal business processes which include:

- One department may be optimized at the expense of another
- Lack of time to focus on improving business process
- Lack of recognition of the extent of the problem
- Lack of training
- People involved use the best tool they have at their disposal which is usually Excel to fix problems
- Inadequate infrastructure
- Overly bureaucratic processes
- Lack of motivation

BPR team composition:

Once organization-wide commitment has been secured from all departments involved in the reengineering effort and at different levels, the critical step of selecting a BPR team must be taken. The determinants of an effective BPR team may be summarized as follows:

- Competency of the members of the team, their motivation,
- Their credibility within the organization and their creativity,
- Team empowerment, training of members in process mapping and brainstorming techniques,
- Effective team leadership,
- Proper organization of the team,

- Complementary skills among team members, adequate size, interchangeable accountability, clarity of work approach, and
- Specificity of goals.
- The most effective BPR teams include active representatives from the following work groups: top management, business area responsible for the process being addressed, technology groups, finance, and members of all ultimate process users' groups.

The BPR team should be mixed in depth and knowledge. For example, it may include members with the following characteristics:

- Members who do not know the process at all.
- Members who know the process inside-out.
- Customers, if possible.
- Members representing impacted departments.
- One or two members of the best, brightest, passionate, and committed technology experts.
- Members from outside of the organization

Business needs analysis:

This plan includes the following:

- Identifying specific problem areas.
- Solidifying particular goals and
- Defining business objectives.
- The business needs analysis contributes tremendously to the re-engineering effort by helping the BPR team to prioritize and determine where it should focus its improvements efforts.

6. Explicate about Management Information System (MIS).

Management Information Systems (MIS), are information systems, typically computer based, that are used within an organization. WordNet described an information system as "a system consisting of the network of all communication channels used within an organization".

Definition:

MIS is a computer – based system that optimizes the collection, collation, transfer and presentation of information throughout an organization, through an integrated structure of databases and information flow.

Characteristics of MIS

- MIS supports data processing functions of transaction handling and record keeping.
- MIS uses an integrated database and supports a variety of functional areas.
- MIS provides operational, tactical and strategic levels of organization with timely, structured information.
- MIS is flexible and can adapt to the changing needs of the organization.

Importance of MIS

- As an area of study it is commonly referred to as information technology management.
- The study of information systems is usually a commerce and business administration discipline, and frequently involves software engineering, but also distinguishes itself by concentrating on the integration of computer systems with the aims of the organization.
- The area of study should not be confused with Computer Science which is more theoretical and mathematical in nature or with Computer Engineering which is more engineering.
- In business, information systems support business processes and operations, support decision making, and support competitive strategies.

The major differences between a management information system and a Data Processing system are:

- The integrated database of the MIS enables greater flexibility in meeting the information needs of the management.
- The MIS integrates the information flow between functional areas (accounting, marketing, manufacturing, etc.) whereas data processing systems tend to support a single functional area.
- MIS caters to the information needs of all levels of management whereas data processing systems focus on departmental-level support.

- Management’s information needs are supported on a timelier basis with the MIS (with its on-line query capability) than with a data processing system.

	MIS		DPS(DATA PROCESSING SYSTEM)
1	It uses an integrated database.	1	It does not use an integrated database.
2	It provides greater flexibility to the management	2	It provides no such flexibility
3	Integrates the information flow between functional areas.	3	DPS tends to support a single functional area.
4	Focus on information needs of all levels of management	4	DPS focus on departmental-level support

Advantages:

The following are some of the benefits that can be attained using MISs.

- Companies are able to identify their strengths and weaknesses due to the presence of revenue reports, employees' performance record etc. Identifying these aspects can help a company improve its business processes and operations.
- Giving an overall picture of the company.
- Acting as a communication and planning tool.
- The availability of customer data and feedback can help the company to align its business processes according to the needs of its customers. The effective management of customer data can help the company to perform direct marketing and promotion activities.
- MISs can help a company gain a competitive advantage. Competitive advantage is a firm’s ability to do something better, faster, cheaper, or uniquely, when compared with rival firms in the market.

7. **Describe about Decision support system (DSS).**

Decision Support Systems (DSS)

- Decision support systems (DSS) are interactive software-based systems intended to help managers in decision-making by accessing large volumes of information generated from various related information systems involved in organizational business processes such as office automation system, transaction processing system, etc.
- DSS uses the summary information, exceptions, patterns, and trends using the analytical models. A decision support system helps in decision-making but does not necessarily give a decision itself.
- The decision makers compile useful information from raw data, documents, personal knowledge, and/or business models to identify and solve problems and make decisions.

Programmed and Non-programmed Decisions

- There are two types of decisions - programmed and non-programmed decisions. Programmed decisions are basically automated processes, general routine work, where:
  - These decisions have been taken several times.
  - These decisions follow some guidelines or rules.
- Non-programmed decisions occur in unusual and non-addressed situations, so:
  - It would be a new decision
  - There will not be any rules to follow.
  - These decisions are made based on the available information.
  - These decisions are based on the manger's discretion, instinct, perception and judgment.

Attributes of a DSS

- Adaptability and flexibility
- High level of Interactivity
- Ease of use
- Efficiency and effectiveness
- Complete control by decision-makers
- Ease of development
- Extendibility
- Support for modeling and analysis
- Support for data access
- Standalone, integrated, and Web-based

Characteristics of a DSS

- Support for decision-makers in semi-structured and unstructured problems.

- Support for managers at various managerial levels, ranging from top executive to line managers.
- Support for individuals and groups. Less structured problems often requires the involvement of several individuals from different departments and organization level.
- Support for interdependent or sequential decisions.
- Support for intelligence, design, choice, and implementation.
- Support for variety of decision processes and styles.
- DSSs are adaptive over time

Classification of DSS

- There are several ways to classify DSS. Hoi Apple and Whinstone classifies DSS as follows:
- **Text Oriented DSS:** It contains textually represented information that could have a bearing on decision. It allows documents to be electronically created, revised and viewed as needed.
- Database Oriented DSS: Database plays a major role here; it contains organized and highly structured data.
- **Spreadsheet Oriented DSS:** It contains information in spread sheets that allows create, view, modify procedural knowledge and also instructs the system to execute self-contained instructions. The most popular tool is Excel and Lotus 1-2-3.
- **Solver Oriented DSS:** It is based on a solver, which is an algorithm or procedure written for performing certain calculations and particular program type.
- Rules Oriented DSS: It follows certain procedures adopted as rules.
- Procedures are adopted in rules oriented DSS. Export system is the example.
- **Compound DSS:** It is built by using two or more of the five structures explained above

Components of a DSS

- **Database Management System (DBMS):** To solve a problem the necessary data may come from internal or external database. In an organization, internal data are generated by a system such as TPS and MIS. External data come from a variety of sources such as newspapers, online data services, databases (financial, marketing, human resources).
- **Model Management System:** It stores and accesses models that managers use to make decisions. Such models are used for designing manufacturing facility, analyzing the financial health of an organization, forecasting demand of a product or service, etc.
- **Support Tools**: Support tools like online help; pulls down menus, user interfaces, graphical analysis, error correction mechanism, facilitates the user interactions with the system.

8. **Detail about Executive Information System/Executive Support Systems.**

- Executive support systems are intended to be used by the senior managers directly to provide support to non-programmed decisions in strategic management.
- These information are often external, unstructured and even uncertain. Exact scope and context of such information is often not known beforehand. This information is intelligence based:
  1. Market intelligence
  2. Investment intelligence
  3. Technology intelligence
- A computerized system intended to provide current and appropriate information to support executive decision making for managers using a networked workstation.
- The emphasis is on graphical displays and an easy to use interface that present information from the corporate database.
- They are tools to provide canned reports or briefing books to top-level executives. They offer strong reporting and drill-down capabilities. An early term for a sophisticated data-driven DSS targeted to senior executives.
- Executive information systems (EIS) provide a variety of internal and external information to top managers in a highly summarized and convenient form. EIS are becoming an important tool of top-level control in many organizations.
- They help an executive spot a problem, an opportunity, or a trend

Characteristics of Executive Information Systems

- EIS provide immediate and easy access to information reflecting the key success factors the company and of its units.
- A User-seductive@ interfaces, presenting information through color graphics or video, allow an EIS user to grasp trends at a glance. 3. EIS provide access to a variety of databases, both internal and external, through a uniform interface.
- Both current status and projections should be available from EIS.
- An EIS should allow easy tailoring to the preferences of the particular users or group of users.
- EIS should offer the capability to Adrill down@ into the data.

DSS are primarily used by middle and lower level managers to project the future, EIS's primarily serve the control needs of higher level management.

- EISs primarily assist top management in uncovering a problem or an opportunity. Analysts and middle managers can subsequently use a DSS to suggest a solution to the problem.
- At the heart of an EIS lies access to the data. EISs may work on the data extraction principal, as DSSs do, or they may be given access to the actual corporate databases or data warehouses.
- EISs can reside on personal workstations or servers

Developing EIS

- EIS's should make it easy to track the critical success factors (CSF) of the enterprise, that is, the few vital indicators of the firm's performance.
- With the use of this methodology, executives may define just the few indicators of corporate performance they need. With the drill-down capability, they can obtain more detailed data behind the indicators.
- Strategic business objectives methodology of EIS development takes a company-wide perspective of the strategic business objectives of the firm where the critical businesses are identified and prioritized.
- Then the information needed to support these processes is defined, to be obtained with the EIS that is being planned. Finally, an EIS is developed to report on the CSFs. This methodology avoids the frequent pitfall of aligning an EIS too closely to a particular sponsor.

Advantages of ESS

- Easy for upper level executive to use
- Ability to analyze trends
- Augmentation of managers' leadership capabilities
- Enhance personal thinking and decision-making
- Contribution to strategic control flexibility
- Enhance organizational competitiveness in the market place
- Instruments of change
- Increased executive time horizons
- Better reporting system
- Improved mental model of business executive
- Help improve consensus building and communication

Disadvantage of ESS

- Functions are limited
- Hard to quantify benefits
- Executive may encounter information overload
- System may become slow
- Difficult to keep current data
- May lead to less reliable and insecure data
- Excessive cost for small company

9. Explain in Detail about OLAP (Online Analytical Processing).

- OLAP is an acronym for On Line Analytical Processing. It is an approach to quickly provide the answer to analytical queries that are dimensional in nature.
- It is part of the broader category business intelligence, which also includes Extract transform load (ETL), relational reporting and data mining.
- The typical applications of OLAP are in business reporting for sales, marketing, management reporting, business performance management (BPM), budgeting and forecasting, financial reporting and similar areas.
- The term OLAP was created as a slight modification of the traditional database term OLTP (On Line Transaction Processing).
- Databases configured for OLAP employ a multidimensional data model, allowing for complex analytical and ad-hoc queries with a rapid execution time.
- Nigel Pendse has suggested that an alternative and perhaps more descriptive term to describe the concept of OLAP is Fast Analysis of Shared Multidimensional Information (FASMI). They borrow aspects of navigational databases and hierarchical databases that are speedier than their relational

Functionality

- OLAP takes a snapshot of a set of source data and restructures it into an OLAP cube. The queries can then be run against this. It has been claimed that for complex queries OLAP can produce an answer in around 0.1% of the time for the same query on OLTP relational data.
- The cube is created from a star schema or snowflake schema of tables. At the center is the fact table which lists the core facts which make up the query. Numerous dimension tables are linked to the fact tables. These tables indicate how the aggregations of relational data can be analyzed. The

number of possible aggregations is determined by every possible manner in which the original data can be hierarchically linked.

- For example a set of customers can be grouped by city, by district or by country; so with 50 cities, 8 districts and two countries there are three hierarchical levels with 60 members.
- These customers can be considered in relation to products; if there are 250 products with 20 categories, three families and three departments then there are 276 product members.
- With just these two dimensions there are 16,560 (276 * 60) possible aggregations. As the data considered increases the number of aggregations can quickly total tens of millions or more.
- The calculation of the aggregations AND the base data combined make up an OLAP cube, which can potentially contain all the answers to every query which can be answered from the data (as in Gray, Bosworth, Layman, and Pirahesh, 1997). Due to the potentially large number of aggregations to be calculated, often only a predetermined number are fully calculated while the remainder are solved on demand.

Types of OLAP

- **Multidimensional or MOLAP:** MOLAP is the classic form of OLAP and is sometimes referred to as just OLAP. MOLAP uses database structures that are generally optimal for attributes such as time period, location, product or account code. The way that each dimension will be aggregated is defined in advance by one or more hierarchies.
- **Relational or ROLAP:** ROLAP works directly with relational databases. The base data and the dimension tables are stored as relational tables and new tables are created to hold the aggregated information. Depends on a specialized schema design.
- **Hybrid or HOLAP:** There is no clear agreement across the industry as to what constitutes "Hybrid OLAP", except that a database will divide data between relational
- and specialized storage.• For example, for some vendors, a HOLAP database will use relational tables to hold the larger quantities of detailed data, and use specialized storage for at least some aspects of the smaller quantities of more- aggregate or less-detailed data

Comparison:

- Each type has certain benefits, although there is disagreement about the specifics of the benefits between providers.

- Some MOLAP implementations are prone to database explosion. Database explosion is a phenomenon causing vast amounts of storage space to be used by MOLAP databases when certain common conditions are met: high number of dimensions, pre-calculated results and sparse multidimensional data.
- The typical mitigation technique for database explosion is not to materialize all the possible aggregation, but only the optimal subset of aggregations based on the desired performance vs. storage trade off.
- MOLAP generally delivers better performance due to specialized indexing and storage optimizations. MOLAP also needs less storage space compared to ROLAP because the specialized storage typically includes compression techniques.
- ROLAP is generally more scalable. However, large volume pre-processing is difficult to implement efficiently so it is frequently skipped. ROLAP query performance can therefore suffer.
- Since ROLAP relies more on the database to perform calculations, it has more limitations in the specialized functions it can use.
- HOLAP encompasses a range of solutions that attempt to mix the best of ROLAP and MOLAP. It can generally pre-process quickly, scale well, and offer good function support.

Advantages of MOLAP

- Fast query performance due to optimized storage, multidimensional indexing and caching.
- Smaller on-disk size of data compared to data stored in relational database due to compression techniques.
- Automated computation of higher level aggregates of the data.
- It is very compact for low dimension data sets.
- Array models provide natural indexing.
- Effective data extraction achieved through the pre-structuring of aggregated data.

Disadvantages of MOLAP

- Within some MOLAP Solutions the processing step (data load) can be quite lengthy, especially on large data volumes. This is usually remedied by doing only incremental processing, i.e., processing only the data which have changed (usually new data) instead of reprocessing the entire data set.
- Some MOLAP methodologies introduce data redundancy

10. Discuss in detail about Supply chain management (SCM).

Supply chain management (SCM)

- Supply chain management is the systemic, strategic coordination of the traditional business functions and tactics across these business functions - both within a particular company and across businesses within the supply chain- all coordinated to improve the long-term performance of the individual companies and the supply chain as a whole.
- In a traditional manufacturing environment, supply chain management meant managing movement and storage of raw materials, work-in-progress inventory, and finished goods from point of origin to point of consumption.
- It involves managing the network of interconnected smaller business units, networks of channels that take part in producing a merchandise of a service package required by the end users or customers. With businesses crossing the barriers of local markets and reaching out to a global scenario, SCM is now defined as:
- Design, planning, execution, control, and monitoring of supply chain activities with the objective of creating net value, building a competitive infrastructure, leveraging worldwide logistics, synchronizing supply with demand and measuring performance globally.
- SCM consists of:
  - Operations management
  - Logistics
  - Procurement
  - information technology
  - integrated business operations

Objectives of SCM

- To decrease inventory cost by more accurately predicting demand and scheduling production to match it.
- To reduce overall production cost by streamlining production and by improving information flow.
- To improve customer satisfaction.

Features of SCM

- Integrated behavior
- Mutually sharing information
- Mutually sharing channel and risk rewards

- Focus on serving customers
- Co-operation
- Partnership to build maintain long term relationship
- Integration of process

SCM Processes

- Customer Relationship Management
- Customer Service Management
- Demand Management
- Customer Order Fulfillment
- Manufacturing Flow Management
- Procurement Management
- Product Development and Commercialization
- Returns Management

Advantages of SCM

SCM have multi-dimensional advantages:

- To the suppliers:
    - Help in giving clear-cut instruction
    - Online data transfer reduce paper work
- Inventory Economy:
    - Low cost of handling inventory
    - Low cost of stock outage by deciding optimum size of replenishment orders
    - Achieve excellent logistical performance such as just in time
- Distribution Point:
    - Satisfied distributor and whole seller ensure that the right products reach the right place at right time
    - Clear business processes subject to fewer errors
    - Easy accounting of stock and cost of stock
- Channel Management:
    - Reduce total number of transactions required to provide product assortment
    - Organization is logically capable of performing customization requirements
- Financial management:

- Low cost
- Realistic analysis

- Operational performance:
  - It involves delivery speed and consistency.
- External customer:
  - Conformance of product and services to their requirements
  - Competitive prices
  - Quality and reliability
- To employees and internal customers:
  - Teamwork and cooperation
  - Efficient structure and system

11. Illustrate about Customer relationship management (CRM).

- CRM is an enterprise application module that manages a company's interactions with current and future customers by organizing and coordinating, sales and marketing, and providing better customer services along with technical support.
- Atul Parvatiyar and Jagdish N. Sheth provide an excellent definition for customer relationship management in their work titled.
- 'Customer Relationship Management: Emerging Practice, Process, and Discipline': Customer Relationship Management is a comprehensive strategy and process of acquiring, retaining, and partnering with selective customers to create superior value for the company and the customer.
- It involves the integration of marketing, sales, customer service, and the supply-chain functions of the organization to achieve greater efficiencies and effectiveness in delivering customer value.

Why CRM?

- To keep track of all present and future customers.
- To identify and target the best customers.
- To let the customers know about the existing as well as the new products and services.
- To provide real-time and personalized services based on the needs and habits of the existing customers.
- To provide superior service and consistent customer experience.
- To implement a feedback system.

CHARACTERISTICS OF CRM

- Builds a database that describes the customers and the relationship they hold with the company.
- Database: a collection of information that is organized in a way that allows it to be easily accessed, managed and updated.
- Provides enough detail so that the company can offer the client the product/service that matches their need the best.
- May contain information about their past purchases, who is involved with the account and a summary of all conversations.

GOALS OF CRM

- Make sure the primary goal (ie: customer service) is standard in the software. Customization is very expensive.
- Be able to integrate smoothly and flawlessly.
- Make sure the program doesn't offer more then you need.
- Compare with other companies of equivalently the same size who have purchased the same CRM software and examine their results.

BENEFITS OF CRM

- Help marketing departments identify and target their best customers, manage campaigns as well as discover qualified leads.
- Qualified Leads: prospects who seem most likely to buy because of some information known about them.
- Improve sales and streamline existing processes.
- Form individualized relationships with customers.
- Give employees information needed to improve customer service and also to better understand customer needs.
- Tracks all points of contact between a customer and the company.
- CRM quickly manages the scheduling of follow-up sales calls to assess the satisfaction of customers and their repurchase probabilities (when and how much).
- Identifying prospects and helps them become customers.
- Closing sales more effectively and efficiently.

- Allowing customers to perform business transactions quickly and easily.
- Providing better service and support following a sale.

CUSTOMER SERVICE

- Helps make call centers more efficient.
- Aids in cross and up selling products.
- Cross Selling: Provide additional products/services.
- Up Selling: Upgrade existing products/services.
- Helps sales staff close deals faster.
- Simplifies marketing and sales processes.
- Allows companies to discover new customers.

COMPARISON OF CRM AND BI

- CRM is closely related to business intelligence because both methods involve using technology to gather, analyze and organize data in order to develop relevant information.
- CRM is just a more specific form of BI that concentrates strictly on customer's behaviors and actions, for both past and future information.

Advantages of CRM

- Provides better customer service and increases customer revenues.
- Discovers new customers.
- Cross-sells and up-sells products more effectively.
- Helps sales staff to close deals faster.
- Makes call centers more efficient.
- Simplifies marketing and sales processes.

Disadvantages of CRM

- Sometimes record loss is a major problem.
- Overhead costs.
- Giving training to employees is an issue in small organizations.